Chapter 6: Uniform Circular Motion and Gravitation

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PHY201
Lecture 6

Part I

Rotation Angle and Angular Velocity

Outline of Part I: Rotation Angle and Angular Velocity

Uniform Circular Motion

Tangential Velocity

Uniform Circular Motion (UCM)

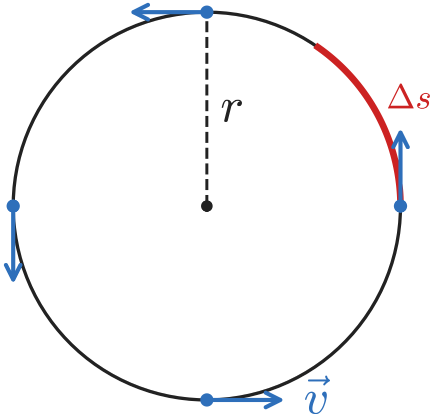
Uniform circular motion is motion along a **circular path** at **constant speed**.

- Speed $v = |\vec{v}|$ is constant.
- The **direction** of $\vec{v}$ changes continuously — so acceleration is **not** zero.

Everyday Examples

- Car around a curve at constant speed
- Earth orbiting the Sun
- Clothes in a washing machine drum

Constant speed does **not** mean constant velocity. Direction change $\Rightarrow$ acceleration.



The Radian and Rotation Angle

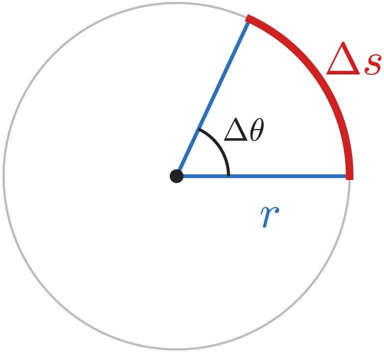
The **radian** (rad) is the SI unit for angles. It is the ratio of arc length to radius:

$$\Delta\theta = \frac{\Delta s}{r}$$

where Δs is the arc length and r is the radius of the circular path.

- 1 revolution $= 2\pi$ rad $= 360°$
- $\Delta\theta$ is dimensionless (rad $=$ m/m)
- Full circle: $\Delta s = 2\pi r \Rightarrow \Delta\theta = 2\pi$ rad ✓

	Degrees	**Radians**
Quarter turn	90°	$\pi/2$
Half turn	180°	π
Full turn	360°	2π



Angular Velocity

Angular Velocity (ω) — SI Unit: $\frac{\text{rad}}{\text{s}}$

Angular velocity is the rate of change of rotation angle:

$$\omega = \frac{\Delta\theta}{\Delta t}$$

where $\Delta\theta$ is the rotation angle (rad) and Δt is the elapsed time (s).

- ω is the angular analogue of linear velocity v
- A clock's minute hand: $\omega = 2\pi\,\text{rad}/3600\,\text{s} = 1.75 \times 10^{-3}\,\frac{\text{rad}}{\text{s}}$
- A CD at 500 rpm: $\omega = 500 \times 2\pi/60 = 52.4\,\frac{\text{rad}}{\text{s}}$

Converting angular velocity to revolutions per second:

$$\omega\,(\text{rev/s}) = \omega\,(\text{rad/s}) \times \frac{1\,\text{rev}}{2\pi\,\text{rad}}$$

Example: Alice on a Merry-Go-Round

Rotation Angle and Angular Velocity — (cp2e_ch6)

Alice sits $r = 3.00\,\mathrm{m}$ from the center of a merry-go-round.
She travels a total arc length of $\Delta s = 60.0\,\mathrm{m}$ during the ride, which lasts $\Delta t = 15.0\,\mathrm{s}$.
Find: (a) her rotation angle, (b) her average angular velocity.

$$r = 3.00\,\mathrm{m},\ \Delta s = 60.0\,\mathrm{m},\ \Delta t = 15.0\,\mathrm{s}$$

$$\text{(a)} \quad \Delta\theta = \frac{\Delta s}{r} = \frac{60.0}{3.00}$$

$$\boxed{\Delta\theta = 20.0\,\mathrm{rad}}$$

$$\text{(b)} \quad \omega = \frac{\Delta\theta}{\Delta t} = \frac{20.0}{15.0}$$

$$\boxed{\omega = 1.33\,\frac{\mathrm{rad}}{\mathrm{s}}}$$

In revolutions per second:

$$\omega = 1.33\,\frac{\mathrm{rad}}{\mathrm{s}} \times \frac{1\,\mathrm{rev}}{2\pi\,\mathrm{rad}}$$
$$= 0.212\,\mathrm{rev/s}$$

Outline of Part I: Rotation Angle and Angular Velocity

Uniform Circular Motion

Tangential Velocity

Tangential Velocity

Tangential (Linear) Speed (v) — SI Unit: $\frac{\mathrm{m}}{\mathrm{s}}$

The speed of a point on a rotating object is related to the angular velocity by:

$$v = r\omega$$

where r is the distance from the axis of rotation and ω is the angular velocity.

- At the center $(r = 0)$: $v = 0$ — no linear speed.
- Farther from the axis $\Rightarrow$ faster linear speed.
- On a merry-go-round: outer seats move faster than inner seats, even though ω is the same for all riders.

$$v = r\omega \qquad \Longleftrightarrow \qquad \omega = \frac{v}{r}$$

Tangential velocity is always **perpendicular** to the radius.

Example: Paint Roller

Rotation Angle, Speed, and Angular Velocity — (cp2e_ch6)

A paint roller has radius $r = 5.00\,\text{cm}$.
It rolls across a wall covering a painted length $\Delta s = 1.60\,\text{m}$ in $\Delta t = 2.00\,\text{s}$.
Find: (a) the rotation angle, (b) the tangential speed, (c) the angular velocity.

$$r = 0.0500\,\text{m},\ \Delta s = 1.60\,\text{m},\ \Delta t = 2.00\,\text{s}$$

$$\text{(a)}\quad \Delta\theta = \frac{\Delta s}{r} = \frac{1.60}{0.0500} = 32.0\,\text{rad}$$

$$\text{(b)}\quad v = \frac{\Delta s}{\Delta t} = \frac{1.60}{2.00} = 0.800\,\frac{\text{m}}{\text{s}}$$

$$\text{(c)}\quad \omega = \frac{\Delta\theta}{\Delta t} = \frac{32.0}{2.00}$$

$$\boxed{\omega = 16.0\,\frac{\text{rad}}{\text{s}}}$$

Check using $v = r\omega$:

$$v = (0.0500)(16.0)$$
$$= 0.800\,\frac{\text{m}}{\text{s}}\,\checkmark$$

Check Your Understanding

A fan blade completes 12.0 revolutions in $\Delta t = 4.00\,\text{s}$.

Find: (a) the rotation angle in radians, (b) the average angular velocity.

Come to lecture to see the answer.

Check Your Understanding

A Ferris wheel of radius $r = 12.0\,\mathrm{m}$ rotates at $\omega = 0.500\,\frac{\mathrm{rad}}{\mathrm{s}}$.

Find the tangential speed of a rider.

Come to lecture to see the answer.

Part II

Centripetal Acceleration and Force

Outline of Part II: Centripetal Acceleration and Force

Centripetal Acceleration

Centripetal Force

Acceleration in Uniform Circular Motion

In UCM, speed is constant but **direction** changes continuously.

Recall from Ch. 2–3: $\vec{a} = \Delta\vec{v}/\Delta t$.
Even when $|\vec{v}|$ is constant, changing direction means $|\Delta\vec{v}| > 0$, so $|\vec{a}| > 0$.

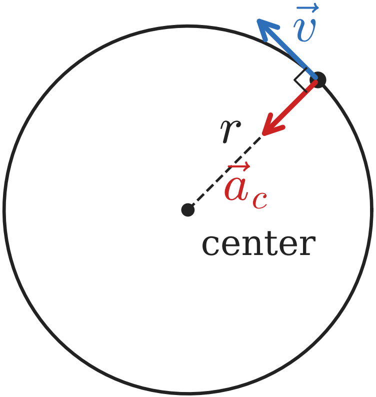
Centripetal Acceleration (a_c) — SI Unit: $\frac{\text{m}}{\text{s}^2}$

In UCM, the acceleration is directed **toward the center** of the circle:

$$a_c = \frac{v^2}{r} = r\omega^2$$

"Centripetal" means **center-seeking**.

a_c is **not** zero just because speed is constant.
Speed and velocity are different.



Check Your Understanding

A motorcycle goes around a circular track of radius $r = 200\,\mathrm{m}$ at $v = 30.0\,\frac{\mathrm{m}}{\mathrm{s}}$.

Find the centripetal acceleration.

Come to lecture to see the answer.

Outline of Part II: Centripetal Acceleration and Force

Centripetal Acceleration

Centripetal Force

Centripetal Force

Centripetal Force $\left(\vec{F}_c\right)$ — SI Unit: N

By Newton's Second Law ($\vec{F} = m\vec{a}$, from Ch. 4), the net inward force is:

$$\vec{F}_c = m\vec{a}_c \qquad \Longrightarrow \qquad F_c = \frac{mv^2}{r} = mr\omega^2$$

F_c is directed **toward the center** of the circular path.

☠️ **Centripetal force is not a new kind of force.**
It is the **role played** by an existing force (e.g. gravity, friction, tension, ...).

💡 Think of centripetal force like the role of "Hamlet" in a play.
Hamlet is not a person — it is a character that an actor must play.
Centripetal force is not a force — it is a role that a real force must occupy.

What Plays the Role of Centripetal Force?

Situation	**Force acting as F_c**	**Direction**
Satellite in orbit	Gravity F_g	Toward Earth
Car rounding a curve	Static friction f_s	Toward center
Ball on a string	Tension T	Toward center
Car on a banked curve	Normal force component F_N	Toward center
Electron around nucleus	Electromagnetic force	Toward nucleus

$$\sum \vec{F}_{\text{inward}} = F_c = \frac{mv^2}{r}$$

Apply Newton's 2nd Law in the radial (inward) direction — whatever forces point inward are the centripetal force.

Example: Maximum Speed in a Curve

Car Rounding a Flat Curve — (cp2e_ch6)

A car of mass $m = 1200\,\text{kg}$ rounds a flat circular curve of radius $r = 100\,\text{m}$. The coefficient of static friction between tires and road is $\mu_s = 0.600$. Find the maximum speed the car can take the curve without skidding.

$$\sum F_y = 0: \quad N = mg$$

Friction is F_c: $\quad f_s \leq \mu_s N$

$$\frac{mv^2}{r} \leq \mu_s mg$$

$$v^2 \leq \mu_s gr$$

$$v_{\text{max}} = \sqrt{\mu_s gr}$$

$$= \sqrt{(0.600)(9.80)(100)}$$

$$\boxed{v_{\text{max}} = 24.2\,\frac{\text{m}}{\text{s}}}$$

$24.2\,\frac{\text{m}}{\text{s}} \approx 87.1\,\frac{\text{km}}{\text{h}}$

Mass cancels — v_{max} is independent of the car's mass.

- $v \leq \sqrt{\mu_s gr}$: higher μ_s (dry road) $\Rightarrow$ higher safe speed
- Larger r (gentle curve) $\Rightarrow$ higher safe speed

Check Your Understanding

A 1500 kg car rounds a flat curve of radius $r = 80.0\,\mathrm{m}$ at $v = 18.0\,\frac{\mathrm{m}}{\mathrm{s}}$.

What minimum coefficient of static friction is needed?

Come to lecture to see the answer.

Check Your Understanding

For each situation, state what real force acts as the centripetal force:

(a) The Moon orbiting the Earth.
(b) A ball tied to a string swinging in a horizontal circle.
(c) A race car on a steeply banked curve (no friction).

Come to lecture to see the answer.

Part III

Fictitious Forces and Non-Inertial Frames

Outline of Part III: Fictitious Forces and Non-Inertial Frames

Fictitious Forces

Fictitious Forces and Non-Inertial Frames

- $\vec{F}_{net} = ma$ works cleanly in *inertial* reference frames (where $\Delta\vec{v} = 0$).
- Reference frames that accelerate are called non-inertial reference frames.
- Objects in non-inertial frames appear to accelerate even when $F_{net} = 0$.

The Centrifugal "Force"

- Rotating frames are non-inertial frames.
- Riders on a merry-go-round feel as though they are pushed outward.
- Nothing actually exerts an outward force.
- The only real force is the centripetal force; the outward "centrifugal force" appears since the observer is in a rotating frame.

Real forces have **identifiable agents**: gravity (Earth), friction (surface), tension (rope).

Fictitious forces have **no agent** — they disappear when you switch to an inertial frame.

Check Your Understanding

A passenger feels pushed to the outside of a curve when a bus turns.

Is this a real outward force on the passenger? Explain using Newton's Laws from Ch. 4.

Come to lecture to see the answer.

Part IV

Newton's Universal Law of Gravitation

Outline of Part IV: Newton's Universal Law of Gravitation

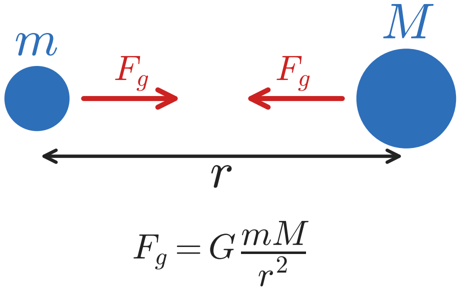
Newton's Universal Law of Gravitation

Every object with mass **attracts** every other object with mass:

$$F_g = G\frac{mM}{r^2}$$

where $G = 6.673 \times 10^{-11}\,\mathrm{N \cdot m^2/kg^2}$ is the **gravitational constant**, m and M are the two masses (kg), and r is the distance between their centers (m).

- $F_g \propto mM$: doubling m or M doubles F_g
- $F_g \propto 1/r^2$: doubling distance reduces F_g by $4\times$ (**inverse square law**)
- Forces on both objects are **equal and opposite** (Newton's 3rd Law)



Weight Revisited: Where Does $w = mg$ Come From?

On Earth's surface, gravity provides weight. Setting $F_g = w = mg$:

$$mg = G\frac{mM_E}{R_E^2} \qquad \Longrightarrow \qquad g = \frac{GM_E}{R_E^2}$$

where $M_E = 5.98 \times 10^{24}\,\text{kg}$ and $R_E = 6.38 \times 10^{6}\,\text{m}$.

calculating g

$$g = \frac{(6.673 \times 10^{-11})(5.98 \times 10^{24})}{(6.38 \times 10^{6})^2}$$

$$g = \frac{3.99 \times 10^{14}}{4.07 \times 10^{13}}$$

$$g = 9.80\,\frac{\text{m}}{\text{s}^2} \checkmark$$

g is not a universal constant — it depends on the planet:

- Moon: $g_{\text{Moon}} \approx 1.62\,\frac{\text{m}}{\text{s}^2}$
- Mars: $g_{\text{Mars}} \approx 3.71\,\frac{\text{m}}{\text{s}^2}$
- Jupiter: $g_{\text{Jup.}} \approx 24.8\,\frac{\text{m}}{\text{s}^2}$

Example: Earth–Moon Gravitational Force

Force Between Earth and Moon — (cp2e_ch6)

$M_E = 5.98 \times 10^{24}\,\text{kg}, \quad M_{\text{Moon}} = 7.35 \times 10^{22}\,\text{kg}, \quad r = 3.84 \times 10^{8}\,\text{m}$

Find the gravitational force between Earth and the Moon.

$$F_g = G\frac{M_E M_{\text{Moon}}}{r^2}$$

$$= \frac{(6.673 \times 10^{-11})(5.98 \times 10^{24})(7.35 \times 10^{22})}{(3.84 \times 10^{8})^2}$$

$$= \frac{(6.673)(5.98)(7.35) \times 10^{35}}{(3.84)^2 \times 10^{16}}$$

$$\boxed{F_g = 1.99 \times 10^{20}\,\text{N}}$$

1.99×10^{20} N is enormous — yet the Moon is in free fall (orbiting) around Earth, not accelerating away or crashing in.

This force provides the **centripetal force** for the Moon's orbit.

Check Your Understanding

Calculate the gravitational force between the Earth ($M_E = 5.98 \times 10^{24}\,\text{kg}, \;\; R_E = 6.38 \times 10^{6}\,\text{m}$) and a $70.0\,\text{kg}$ person standing on its surface.

Compare to $w = mg$.

Come to lecture to see the answer.

Part V

Satellites and Kepler's Laws

Outline of Part V: Satellites and Kepler's Laws

Orbital Speed and Satellites

Kepler's Laws

Orbital Speed: Gravity as Centripetal Force

Orbital Speed (v_{orb})

For a satellite in circular orbit at radius r, gravity provides the centripetal force:

$$F_g = F_c \qquad \Longrightarrow \qquad G\frac{mM}{r^2} = \frac{mv^2}{r}$$

Solving for v (mass of satellite m cancels!):

$$v_{\text{orb}} = \sqrt{\frac{GM}{r}}$$

The orbital speed is **independent of the satellite's mass**. A bowling ball and a space station at the same altitude orbit at the same speed.

Example: ISS Orbital Speed

International Space Station — (cp2e_ch6)

The ISS orbits at altitude $h = 400\,\text{km}$ above Earth's surface.
$M_E = 5.98 \times 10^{24}\,\text{kg}, \quad R_E = 6.38 \times 10^6\,\text{m}$
Find the orbital speed.

$$\begin{aligned} r &= R_E + h = 6.38 \times 10^6 + 4.00 \times 10^5 \\ &= 6.78 \times 10^6\,\text{m} \end{aligned}$$

$$\begin{aligned} v_{\text{orb}} &= \sqrt{\frac{GM_E}{r}} \\ &= \sqrt{\frac{(6.673 \times 10^{-11})(5.98 \times 10^{24})}{6.78 \times 10^6}} \\ &= \sqrt{5.88 \times 10^7} \end{aligned}$$

$$\boxed{v_{\text{orb}} = 7670\,\frac{\text{m}}{\text{s}} \approx 27\,600\,\frac{\text{km}}{\text{h}}}$$

The ISS circles Earth 16 times per day.

Note: as r increases (higher orbit), v_{orb} **decreases**. Higher orbits are slower but have longer periods.

Outline of Part V: Satellites and Kepler's Laws

Orbital Speed and Satellites

Kepler's Laws

Kepler's Three Laws of Planetary Motion

Kepler's Laws (1619)

1st Law Planets travel in **elliptical** orbits with the Sun at one focus.

2nd Law A line from the Sun to a planet sweeps out **equal areas in equal times**.
(A planet moves *faster* when closer to the Sun, *slower* when farther.)

3rd Law The square of the orbital period is proportional to the cube of the orbital radius:

$$\frac{T_1^2}{T_2^2} = \frac{r_1^3}{r_2^3}$$

Kepler found these laws empirically from astronomical data. Newton later **derived** them from his Law of Universal Gravitation — a powerful confirmation.

Kepler's Third Law

Kepler's Third Law

For any two objects orbiting the same central body:

$$\frac{T_1^2}{T_2^2} = \frac{r_1^3}{r_2^3} \qquad \Longleftrightarrow \qquad T_2 = T_1 \left(\frac{r_2}{r_1}\right)^{3/2}$$

Planet	T (yr)	r (AU)
Mercury	0.241	0.387
Earth	1.00	1.00
Mars	1.88	1.52
Jupiter	11.9	5.20
Saturn	29.5	9.54

1 AU = mean Earth–Sun distance $= 1.496 \times 10^{11}$ m

Kepler's 3rd law applies to any set of satellites orbiting the **same** central body: moons of Jupiter, satellites of Earth, exoplanets, etc.

Example: Moons of Mars

Kepler's Third Law — (cp2e_ch6)

Mars has two moons. Moon ① orbits at $r_1 = 1.00 \times 10^6$ m with period $T_1 = 1.00$ yr.
Moon ② orbits at $r_2 = 2.00 \times 10^6$ m.
Find the period T_2 of Moon ②.

$$r_1 = 1.00 \times 10^6\,\text{m},\ \ r_2 = 2.00 \times 10^6\,\text{m}$$

$$T_1 = 1.00\,\text{yr}$$

$$T_2 = T_1 \left(\frac{r_2}{r_1}\right)^{3/2} = (1.00)\left(\frac{2.00}{1.00}\right)^{3/2}$$

$$T_2 = (1.00)\,(2.00)^{3/2} = (1.00)(2.83)$$

$$\boxed{T_2 = 2.83\,\text{yr}}$$

Doubling the orbital radius increases the period by $2^{3/2} = 2\sqrt{2} \approx 2.83\times$.

Larger orbit $\Rightarrow$ longer period (slower speed *and* longer path).

Check Your Understanding

Mars orbits the Sun at $r_{\text{Mars}} = 1.52\,\text{AU}$ with period $T_{\text{Mars}} = 1.88\,\text{yr}$.

Jupiter orbits at $r_{\text{Jup}} = 5.20\,\text{AU}$.

Find Jupiter's orbital period.

Come to lecture to see the answer.

Check Your Understanding

As a satellite moves to a **higher** orbit (larger r):

(a) Does its orbital speed increase or decrease?

(b) Does its orbital period increase or decrease?

Explain both answers using the equations from today.

Come to lecture to see the answer.

Next Lecture

Chapter 7: Work, Energy, and Energy Resources

- Work: $W = \vec{F} \cdot \vec{d} = Fd\cos\theta$
- Kinetic energy: $KE = \frac{1}{2}mv^2$
- Potential energy and conservation of energy
- Power: $P = W/t$

- Energy is conserved — it changes form but is never created or destroyed.
- Work–energy theorem connects Ch. 4–6 force concepts to a new scalar framework.